

Polynomial Degree 1 vs Degree 2

A reproducible model comparison for the Kaggle House Prices dataset

BEST DEGREE	DEGREE 1
Selection rule	Lowest five-fold cross-validated RMSE

Five-fold cross-validation

Degree	Terms	CV MAE	CV RMSE	CV R2
1 - selected	16	\$17,661.62	\$25,784.67	0.8925
2	152	\$18,089.54	\$27,150.27	0.8799

Independent 80/20 holdout

Degree	Holdout MAE	Holdout RMSE	Holdout R2
1 - selected	\$17,957.65	\$24,450.04	0.8918
2	\$17,464.30	\$24,112.02	0.8947

Degree 2 is narrowly better on this one holdout split, but degree 1 wins across five folds. The multi-fold result is the stronger basis for model selection because it is less sensitive to a favorable partition.

Why Degree 1 Wins

Evidence	Degree 1	Degree 2	Interpretation
CV RMSE	\$25,784.67	\$27,150.27	Degree 1 lowers average large-error penalty by \$1,365.60.
CV MAE	\$17,661.62	\$18,089.54	Degree 1 lowers average absolute error by \$427.92.
CV R2	0.8925	0.8799	Degree 1 explains more variance across folds.
Complexity	16 terms	152 terms	Degree 1 uses about 89% fewer transformed terms.

Interpretation

The disagreement between holdout and cross-validation is not an error. It indicates that degree 2 can perform well on a particular split but is less stable across different validation samples. Its 152 terms create more flexibility, which also increases variance and sensitivity to the training partition.

Degree 1 is still polynomial regression

`PolynomialFeatures(degree=1)` produces first-order terms only and is mathematically equivalent to multiple linear regression after preprocessing. It belongs to the same tested polynomial family. Degree 2 adds squares and interactions, but those additions do not improve average unseen-data performance here.

Decision

Use degree 1 for the final fitted model, Streamlit predictions, and the primary Kaggle-format submission. Keep the degree-2 results and submission as comparison artifacts so the decision remains auditable.

Reproducible Comparison Design

Stage	Applied equally to both degrees
Rows	1,458 training rows after the same documented two-row outlier rule
Inputs	20 raw numeric columns transformed into 16 model features
Missing values	Median imputation learned only from each training fold
Transformation	PolynomialFeatures with include_bias=False, then StandardScaler
Target	log1p(SalePrice), converted back to dollars with expm1
Validation	Five shuffled folds with random_state=42 plus one 80/20 holdout
Selection	Lowest mean five-fold CV RMSE

Repository behavior after selection

The training script fits and evaluates both degrees, saves both comparison tables, creates one submission per degree, and copies the selected degree-1 predictions to polynomial_submission.csv. The Streamlit app reads best_degree from metrics.json and fits the selected model. The notebook reproduces the same decision process.

Honest limitation

This conclusion applies to the current data preparation, feature set, outlier rule, target transformation, and validation design. A materially different feature set could change the preferred degree. The project intentionally avoids advanced models and higher polynomial degrees.

Final recommendation: Degree 1

It delivers better average validation performance with far fewer terms, making it the more accurate, stable, and parsimonious choice for this repository.